

$$H A) 1) a) A = \begin{pmatrix} 0 & 1 \\ 7 & -4 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 3 \end{pmatrix}$$

$$b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$G(s) = c^T \underbrace{(sE - A)^{-1}} b + d$$

$$\begin{pmatrix} s & -1 \\ -7 & s+4 \end{pmatrix}^{-1} = \frac{1}{s(s+4) - 7} \begin{pmatrix} s+4 & 1 \\ 7 & s \end{pmatrix}$$

$$-||- \begin{pmatrix} s+4 & 1 \\ 7 & s \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = -||- \begin{pmatrix} s+4+2 \\ 7+2s \end{pmatrix}$$

$$(1 \quad s) -||- \begin{pmatrix} s+4 \\ 2s+7 \end{pmatrix} = \frac{7s + 27}{s^2 + 4s - 7} \quad \checkmark$$

$$b) \quad s^2 + (4 + k_2)s + (-7 + k_1)$$

$$HA(2) \quad G(s) = \frac{20(s+2)}{s(s+4)(s+6)}$$

$$M_p = 0,1$$

$$t_s = 2s$$

3. Polstelle Faktor 10

$$D = \frac{-\ln(M_p)}{\sqrt{\pi^2 + \ln^2(M_p)}} = 0,6$$

$$\omega_0 = \frac{4}{Dt_s} = 3,4$$

$$N = s^2 + 2D\omega_0 s + \omega_0^2$$

$$p_{1,2} = -\omega_0 D \pm j \omega_0 \sqrt{1-D^2} = -2 \pm 2,7j$$

$$p_3 = \underbrace{10}_{\text{Faktor}} \cdot \underbrace{(-\omega_0 \cdot D)}_{\text{Realteil}} = -20$$

$$N_{\text{voll}} = (s+20)(s+2-2,7j)(s+2+2,7j)$$

$$= (s+20)(s^2 + 4s + 11,3)$$

$$s^3 + 24s^2 + 226s + 226$$

$$N_{\text{ist}} = s^3 + 10s^2 + 24s + 0$$

$$\Rightarrow s^3 + (10+K_3)s^2 + (24+K_2)s + (K_1)$$

$$\Rightarrow K_1 = 226$$

$$K_2 = 67,3$$

$$K_3 = 14$$

HA) 3)

$$\dot{x} = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} x + \begin{pmatrix} 1 \\ 1 \end{pmatrix} u$$

$$\left. \begin{array}{l} \lambda_1 = -3 \\ \lambda_2 = -4 \end{array} \right\} s^2 + 7s + 12$$

$$Q_s = \begin{pmatrix} 1 & -1 \\ 1 & -2 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}$$

$$q(A) = A^2 + 7A + 12E$$

$$Q_s^{-1} = \begin{pmatrix} 2 & -1 \\ 1 & -1 \end{pmatrix}$$

$$q(A) = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix} + \begin{pmatrix} -7 & 0 \\ 0 & -14 \end{pmatrix} + \begin{pmatrix} 12 & 0 \\ 0 & 12 \end{pmatrix}$$

$$q(A) = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$$

$$k^T = (0 \ 1) Q_s^{-1} q(A)$$

$$k^T = (0 \ 1) \begin{pmatrix} 2 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$$

$$= (0 \ 1) \begin{pmatrix} 6 & -2 \\ 6 & -2 \end{pmatrix} = (6 \ -2)$$